MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8172

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, metal-glass and ceramic
See figure 1
Mounting position: Any
Weight: 6 ounces nominal

ABSOLUTE RATINGS: F1 (CW) = 500 MHz, F1 (pulsed) = 1,500 MHz

Parameter:	Ef	eb	Eb	ec1	Ec1	ec2	Ec2	Ib	ik	Cooling
Unit:	Vac	kv	Vdc	v	Vdc	kv	Vdc	mAdc	a	Note 1
Maximum:										
(Notes 1 & 2)										
Class C Telep:	2.5±5%	---	1,000	---	-250	---	300	200	---	---
Class C Teleg:	2.5±5%	---	1,250	---	-250	---	300	250	---	---
Class C Pulsed:	2.5±5%	7.0	---	-500	----	1.5	---	---	7	---
TEST CONDITIONS:	2.5	---	1,000	---	Adj	---	300	150	---	Note 3

ABSOLUTE RATINGS: F1 (CW) = 500 MHz, F1 (pulsed) =1,500 MHz

Parameter:	tp	Pg1	Pg2	Pp	Pi	tk	T(ceramic seals and anode core)	T(glass seal
Unit:	u s Note 4	W	W	W	W	sec (min)	°C	°C
Maximum:								
(Notes 1 & 2)								
Class C Telep:	---	2	12	165	200	30	250	150
Class C Teleg:	---	2	12	250	312	30	250	150
Class C Pulsed:	5	2	12	250	312	30	250	150
TEST CONDITIONS:	---	---	---	---	---	120	---	---

GENERAL:

Qualification - Required

The activity responsible for the Qualified Products List is the Defense Electronics Supply Center (DESC-EQA), Dayton OH 45444.

(M) Denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
2212	Pulse emission (1)	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ik = 1.5 a; prr = 11.0 ± 1.0; tp = 4,500 µs (min); Ef = 2.25 Vac	0.65	II	ΔIk	---	200	ma
1372	Current division (long pulse, Method A)	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ib = 1.0 a; prr = 11.0 ±1.0; tp = 4,500 µs (min)	0.65	II	egk	8	18	v
					ic1	---	200	ma
					ic2	---	260	ma
1256	Electrode current (screen)		0.65	II	Ic2	-5.0	+3.0	mAdc
1261	Electrode voltage (grid)		0.65	II	Ec1	-32	-45	Vdc
1266	Total grid current	Eb = 2,000 Vdc; Ib = 125 mAdc	0.65	II	Ic1	---	-15	µAdc
1266	Primary grid emission (control)	Ic1 = 70 mAdc; t = 15; anode and screen grid floating	0.65	II	Isg1	---	-25	µAdc
1266	Primary grid emission (screen)	Ec1 = 0; Ic2 = 100 mAdc; t = 15; anode floating	0.65	II	Isg2	---	-250	µAdc
1301	Heater current		0.65	II	If	6.20	7.30	Aac
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	No voltages applied	---	---	---	---	---	---
1036	Bump	Angle = 20°	---	---	---	---	---	---
2214	RF useful output power	Class C amplifier; F = 460 to 490 MHz; Eb = 1,250 Vdc; Ec1 = -90 Vdc; Ec2 = 250 to 300 Vdc; Ic1 = 20 mAdc (max); Eg1/Ib = 250 mAdc; Ef = 2.30 Vac See Note 5)	---	---	Po	145	---	W useful
1331	Direct-interelectrode capacitance	Grounded grid	---	---	Cpk	---	0.010	pF
					Cin	14.5	19.00	pF
					Cout	4.0	4.90	pF
1331	Direct-interelectrode capacitance	Grounded cathode	---	---	Cgp	---	0.050	pF
					Cin	25.0	29.0	pF
					Cout	4.00	4.90	pF
2212	Pulse emission (2)	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ik = 1.5 a; prr = 11.0 ±1.0; tp = 4,500 µs (min); Ef = 2.50 Vac	---	---	ΔIk	---	100	ma

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
----	Life test(1) provisions	Group C; RF useful output power t = 500 hours	---	---	---	---	---	---
----	Life-test(1) end points:							
2212	Pulse emission (2)		---	---	Δik	---	100	ma
1266	Primary grid emission (control)		---	---	Isg1	---	-100	μAdc
1266	Primary grid emission (screen)		---	---	Isg2	---	-250	μAdc
----	Life-test(2)provisions	Group C; Ec1 = Ec2 = Eb = 0; Ef = 2.75 Vac; t = 500 hours	---	---	---	---	---	---
----	Life-test (2) end points:							
1366	Interelement leakage resistance	supply voltage = 500 Vdc; Rs = 2.5 Meg; anode positive See note 6	---	---	Rg1g2	10	---	Meg
					Rg2k	10	---	Meg
					Rg1k	10	---	Meg
(M) 1011	Humidity	See note 7	---	---	---	---	---	---
----	Post-humidity end point:							
(M) 1266	Total grid current		---	---	Ic1	---	-20	μAdc
(M) ----	Pulse-power output	Eb = 7 kv; ib = 6 a; ec2 = 1.2 kv; tp = 5 μs; prr = 1,000; F = 1,200 ±50 MHz See note 7	---	---	po	17	---	kw

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NOTES:

1. Forced-air cooling shall be provided for the base and anode. The maximum seal and anode core temperature ratings shall not be exceeded. At an anode dissipation of 250 watts and an incoming air temperature of 25°C maximum, a minimum airflow of 5.6 cfm at sea level shall pass through the anode cooler. With no voltages applied to the tube, the static pressure drop across the anode cooler with an airflow of 5.6 cfm, at 25°C sea level, is 0.32 inch of water maximum. The pressure drop varies with the amount of escaping air and with the shape and construction of air director. This airflow requirement applies at bias voltages less than 100 Volts and at frequencies less than 500 MHz (CW) and 1,500 MHz (pulsed). Air cooling on the tube shall be increased with increased negative grid bias, increased incoming air temperature, or increased frequency of operation. The stem shall be cooled either by a portion of the anode air or by a separate air supply. The airflow shall be applied before or simultaneously with them.

2. The transit-time heating effect of the cathode shall be compensated by a rdduction in heater voltage after dynamic operation of the tube has started. The back heating is a function of frequency, grid current, grid bias, and duty cycle. For satisfactory life the heater voltage shall be adjusted to the value indicated below. The rated variation of ±5 percent Ef shall apply to the value selected from this table. Conditions of operation are limited to those which indicate a heater voltage of 2.3 Volts or more.

FREQUENCY (MHz)	Ef (Vac)
Up to 400	2.5
401 to 1,000	2.2
1,000 to 1,500	2.0

3. In all electrical tests involving application of heater voltage, forced-air cooling of the tube shall be allowable at the rate of 6.0 cfm, maximum for the base and anode. A separate source shall be used for the base and anode but neither shall shall exceed 6.0 cfm. The cooling air shall not have a temperature less than 20°C, nor an absolute pressure greater than 32 in. Hg.

4. With a prr of 1,000 or less and a tp of 5 µs or less, ib shall be limited to 6.0 amperes. For pulses of longer duration or higher repetition rates, ib shall be reduced in accordance with the data contained on figure 2.

5. Use cavity and circuit as shown on Drawing 224-JAN.

6. This test shall be made 30 minutes after Ef is turned off. Rated airflow shall be maintained during the 30-minute interval. Measurement to be made with General Radio Megometer Model No. 1862C, or equivalent. Unused elements are to be left floating.

(M) 7. This test shall be performed yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tube with a combined acceptance number of one. In the event of failure, the test will be made as a part of Quality Conformance Inspection, Part 2, AQL 6.5, Inspection Level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

(M) 8. Changes from Previous Issue. The margins of this revision are marked with a letter M to indicate where changes from the previous revision have been made. This was done as a convenience only and the Government assumes no liability for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this revision without regard to the marginal notations and the relationship to the previous issue of this document.

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NOTES: -Continued

(M) 9. Reclaimed materials shall be utilized to the maximum extent possible.

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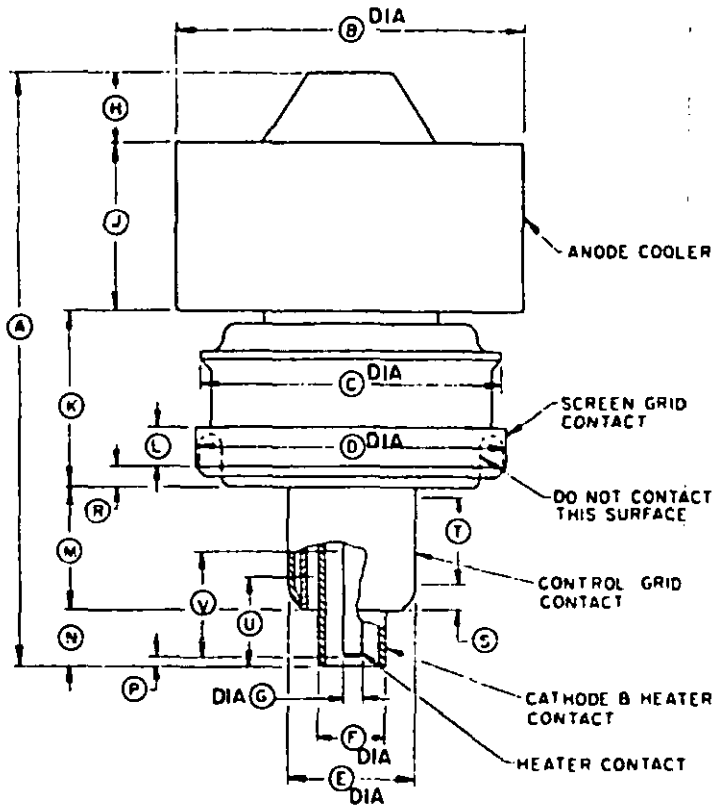
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A		2.750 (69.85)	
B	1.615 (41.02)	1.635 (41.53)	
D	1.417 (35.99)	1.433 (36.40)	
E	.587 (14.91)	.597 (15.16)	
F	.317 (8.05)	.327 (8.31)	
G	.088 (2.24)	.098 (2.49)	
Quality conformance inspection, part 3 (SEE NOTE d)			
C		1.406 (35.71)	
H		.313 (7.95)	
J	.719 (18.26)	.781 (19.84)	b
K	.750 (19.05)	.813 (20.65)	
L	.188 (4.78)		b
M	.500 (12.70)	.578 (14.68)	
N	.234 (5.94)	.266 (6.76)	
P	.031 (.79)	.063 (1.60)	
T	.344 (8.74)		b
U	.406 (10.31)		b
V	.469 (11.91)		b
Reference dimensions			a
R	.094 (2.39)		
S	.125 (3.18)		

NOTES:

a. Reference dimensions are for information only and are not required for inspection purposes.

b. Available contact length.

c. Dome contour shall be such that it will be contained within crosshatched area of template shown on figure 3. Optical comparator techniques are normally used for this purpose.

d. See note 7.

FIGURE 1. Outline drawing of electron tube type 8172.

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